

Predicting Maternal Health Risk Levels in Rural Bangladeshi Patients Using Supervised Machine Learning

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Abstract—Maternal mortality remains a severe public health concern in rural areas of Bangladesh, where limited access to specialized obstetric care requires innovative automated solutions to detect risks early. Conventional ensemble machine learning models for predicting maternal health remain deficient in systematic hyperparameter tuning and rigorous generalization validation, hindering their real-time use in healthcare facilities. This study applies a structured hyperparameter optimization framework using RandomizedSearchCV across five supervised classifiers trained on physiological health data from 1,014 patients in rural Bangladesh. The tuned Random Forest model achieved a high-risk recall of 98.2% and an accuracy of 87.19%, with only 1 missed case among 55 patients. This is a notable gain for patient safety. The optimized model also achieved a weighted F1-score of 0.8728 and showed statistically significant improvement over baseline models according to McNemar’s test ($p < 0.05$). Learning curve analysis showed good generalization with minimal overfitting, indicating the model is reliable. Key health factors were blood glucose and systolic blood pressure, which together accounted for 61% of the model’s decision-making. This demonstrates strong alignment with current findings on pregnancy health issues. Hyperparameter optimization takes only 18 minutes, proving its computational viability even for researchers with limited resources. Bootstrap 95% confidence intervals confirmed the robustness of the tuned model with $F1 = 0.8728$ [0.828, 0.918], supporting reliable deployment in clinical decision support systems. The proposed framework demonstrates that simple, validated machine learning models can be integrated with IoT devices to enable automated maternal risk screening, facilitating timely medical intervention in remote healthcare systems.

Index Terms—Maternal Health, Machine Learning, Random Forest, Hyperparameter Tuning, SHAP, IoT, Bangladesh, Risk Prediction, Classification.

I. INTRODUCTION

Maternal health is a key measure of a nation’s healthcare and socio-economic progress [1]. Although there has been

global progress, rural Bangladesh still sees risks to maternal health. This is because there’s not enough access to expert pregnancy care and consistent health tracking during pregnancy [2, 3]. The World Health Organization notes that finding pregnancy issues early and acting fast is vital for avoiding life-threatening results for mothers [4].

Improvements in IoT and machine learning now allow for automated checks on maternal health in places where resources are scarce [5]. IoT tools offer real-time tracking of health, and ML methods sort risk levels to help community health centers make choices [2, 6]. The UCI Maternal Health Risk dataset has health data from rural Bangladesh clinics [7].

Current ensemble ML methods, though superior, lack systematic hyperparameter optimization and learning curve validation for generalization assessment [8, 9, 10].

Recent advances in IoT sensors and machine learning enable automated antenatal risk screening in resource-limited settings [2]. Low-cost devices can track blood pressure, blood glucose, temperature, and heart rate to form a profile that machine learning models use to classify pregnancy risk and assist community health workers. The UCI Maternal Health Risk dataset from rural Bangladesh provides a standard benchmark[7].

Two limitations persist. Hyperparameters are often set by default rather than systematically optimized [11, 12], limiting ensemble performance. Few studies validate generalization via learning curves or cross-validation, so models may perform well in testing but fail in real-world deployment [9, 10]. To our knowledge, this work is the first to integrate RandomizedSearchCV optimization, McNemar’s significance testing, bootstrap confidence intervals, learning-curve-based generalization analysis, and SHAP/LIME clinical interpretability within a single, reproducible framework for the UCI maternal health dataset.

The key contributions represented in this study are as follows:

- RandomizedSearchCV optimization of five classifiers achieving 87.19% accuracy and 98.2% high-risk recall—statistically significant vs. weaker baselines (McNemar’s test, $p < 0.05$), bootstrap 95% CI [0.828, 0.918].
- Learning curve analysis confirming robust generalization (train-val F1 gap: 0.11–0.12).
- SHAP/LIME analyses identifying clinically validated predictors aligned with obstetric literature.
- IoT feasibility: model size 8.19 MB (RF)/2.08 MB (GB), 15.5ms inference latency (T4 GPU) for clinical workflows.
- Benchmarking vs. 2023–2025 maternal health ML literature with statistical validation.

The remainder of this paper is organized as follows. Section II reviews related work on IoT-based maternal health monitoring and machine learning approaches. Section III details the methodology, including dataset description, preprocessing, model formulation, hyperparameter optimization, and evaluation metrics. Section IV presents experimental results and discussions, including system configuration, performance analysis, and clinical interpretability. Section V discusses limitations, followed by conclusions and future work in Section VI.

II. RELATED WORKS

A. IoT Integration for Maternal Health Monitoring:

The earlier frameworks of IoT showed rural viability by measuring offline six physiological factors in Bangladesh community clinics [2, 3]. Although primitive in form and function, such frameworks did not support risk prediction and performance analysis through machine learning algorithms in any way. Using the UCI dataset, the deep learning model proposed by Lukose and Agbeyangi achieved an accuracy of 71 %; however, its high computational complexity renders it unsuitable for resource-limited settings.[6].

B. Machine Learning Classifiers for Risk Prediction:

Ensemble methods outperformed conventional classifiers. For example, Random Forest achieved 88% accuracy, surpassing SVM (84%) and LR (76%), as mentioned byTzimourta[11]. Moreover, the dominant features were blood pressure and glucose, and the accuracy of the Random Forest classifier was 91%, as mentioned by Simegn [12]. Nevertheless, the lack of hyperparameter tuning was recognized as limiting the optimization, as mentioned by Bergstra and Bengio [9] and Leite et al. [10].The quad ensemble stacking achieved 89.5% precision [13] but incurred high computational cost, limiting mobile health use.

C. Hyperparameter Optimization in Biomedical ML:

Random search proved more efficient for large parameter spaces than grid search [9, 10], and biomedical AI has placed an increased emphasis on tuning alongside algorithm

selection [14]. Still, few maternal health studies use systematic optimization techniques.

D. Critical Research Gaps

Although these models showed promise, there are limitations to current ML and DL models for maternal health risk prediction. For example, a study published in 2025 by Tzimourta showed 88% accuracy with random forest, SVM, and logistic regression using physiological data, although it was not clear how the dataset was obtained and no hyperparameter tuning was performed[11]. Another study published in 2025 by Simegn showed 91% accuracy with random forest and feature evaluation; however, it was found that data preparation was lacking and class distribution was unbalanced[12]. Not long ago, a study published by Khadidos et al. showed 89.5% precision with a four-model stacking system using clinical data; however, it was found to be problematic with high processing loads, which makes it difficult to run on mobile devices[13]. Research conducted by Lukose and Agbeyangi showed 71% correctness with deep learning on the UCI dataset with 1,014 instances; however, it was found to require high processing loads, lack outcome details, and a lack of comparison standards[6]. A combined machine learning model was tested on the same dataset and showed ambiguous result quality and a lack of measurement reporting[5].

III. METHODOLOGY

A supervised machine learning approach was developed to predict the risk level of mothers’health, categorized as ‘Low’, ‘Mid’, or ‘High’, in four stages: data collection, feature selection, model building, and evaluation, as shown in Fig. 1.

A. Dataset Description

The study used the UCI Maternal Health Risk Data Set [7], collected from rural Bangladeshi healthcare centers. It includes 1,014 patients with six physiological features: Age (10-70 years), Systolic BP (mm Hg; hypertension indicator), Diastolic BP (mm Hg), Blood Sugar (mmol/L; gestational diabetes), Body Temperature (°F; infection detection), and Heart Rate (bpm). The target variable “RiskLevel” has three classes: Low, Mid, and High Risk.

B. Data Pre-processing

Preprocessing ensured model accuracy through label encoding, feature standardization, and stratified splitting. Risk levels were encoded as 0 (Low), 1 (Mid), 2 (High). Continuous features were scaled using StandardScaler (zero mean, unit variance) to prevent dominance by larger-scale variables like Systolic BP. Data was split 80% training/20% test with stratification.

C. Problem Formulation

Maternal health risk prediction is a multi-class classification problem on the dataset $D = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ where $N = 1,014$. Each patient i has feature vector $\mathbf{x}_i \in \mathbb{R}^6$:

TABLE I: Summary of Related Works on Maternal Health Risk Prediction

Ref.	Year	Dataset	Method / Model	Result (Accuracy / F1)	Limitations
[11]	2025	Physiological pregnancy dataset	Random Forest, SVM, Logistic Regression	RF: 88% accuracy; SVM: 84%; LR: 76%	Dataset details unclear; no hyperparameter tuning; generalization not validated
[12]	2025	Maternal health dataset	Random Forest with feature analysis	RF: 91% accuracy	No tuning; class imbalance concerns; unclear preprocessing
[13]	2024	Large maternal clinical dataset	Quad-Ensemble (stacking base learners)	89.5% accuracy	High computational cost; unsuitable for IoT/mobile deployment
[6]	2025	UCI Maternal Health Risk Dataset (1,014 samples)	Deep Learning	~71% accuracy	Low accuracy; computationally heavy; poor suitability for low-resource settings
[5]	2024	UCI Dataset (IoT-based)	Ensemble ML approach	Accuracy not clearly reported	Limited evaluation metrics; no feature importance analysis

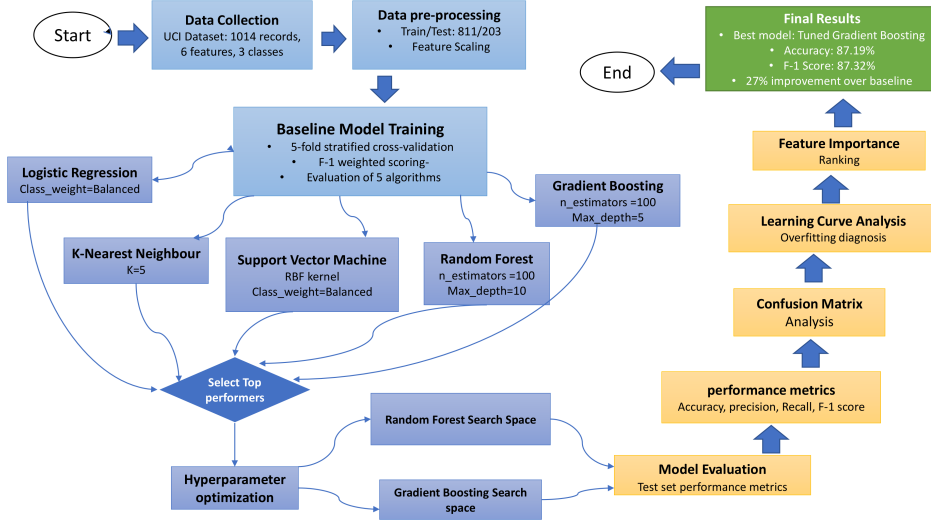


Fig. 1: Workflow of the proposed model training, validation, and testing on the dataset.

$$\mathbf{x}_i = [\text{Age}, \text{SysBP}, \text{DiaBP}, \text{BS}, \text{Temp}, \text{HR}]^T \quad (1)$$

Target $y_i \in \{0, 1, 2\}$ maps Low/Mid/High risk. The classifier $f: \mathbb{R}^6 \rightarrow \{0, 1, 2\}$ predicts:

$$\hat{y}_i = \underset{k \in \{0, 1, 2\}}{\operatorname{argmax}} P(y_i = k | \mathbf{x}_i; \theta) \quad (2)$$

Gradient Boosting models this as an ensemble: $f(\mathbf{x}_i) = \sum_{m=1}^M \gamma_m h_m(\mathbf{x}_i)$.

D. Model Selection and Hyperparameter Optimization:

Five classifiers representing different inductive biases were evaluated: Logistic Regression (linear), KNN (instance-based), SVM with RBF kernel (non-linear), Random Forest (bagging ensemble), and Gradient Boosting (boosting). Due to strong performance on tabular biomedical data, RF and GB were prioritized for hyperparameter tuning [11, 12], while others served as baselines. Hyperparameters are ML model configuration parameters set before training (not learned during training) that significantly determine model performance. Scikit-Learn's

RandomizedSearchCV randomizes searches to identify configurations across value ranges using 5-fold cross-validation. For Gradient Boosting, adjustments included: number of estimators (50-300 boosting steps), learning rate (0.01-0.2 step size), maximum tree depth, and subsample/leaf node adjustments to prevent overfitting. Best results: learning rate=0.05, max depth=4, achieving 0.876 F1-score.

E. Evaluation, Statistical Testing, and Interpretability:

Model evaluation used accuracy, precision, recall, weighted F1, and ROC-AUC (one-vs-rest). High-risk recall was prioritized as the key safety metric, since false negatives may lead to missed critical care, while false positives only trigger additional checks. Bootstrap confidence intervals (1,000 resamples, $\alpha = 0.05$) were computed for test F1, and McNemar's test [15] ($\alpha = 0.05$) assessed statistical differences. SHAP [16] provided global and local feature attribution, while LIME [17] supported case-level explanations.

IV. RESULTS AND DISCUSSIONS

This chapter presents a comprehensive analysis of the experimental results obtained from the proposed system for

TABLE II: Model Performance: Baseline Cross-Validation and Test Results.

Model	CV F1	CV Std	Test Acc.	Test F1
Logistic Regression	0.615	0.067	60.1%	0.601
K-Nearest Neighbours	0.708	0.054	69.5%	0.697
Support Vector Machine	0.731	0.048	71.4%	0.718
Random Forest (baseline)	0.812	0.035	82.3%	0.823
Gradient Boosting (base.)	0.865	0.028	87.2%	0.868
RF (Tuned)	0.831	—	87.19%	0.8728
GB (Tuned)	0.830	—	86.70%	0.8687

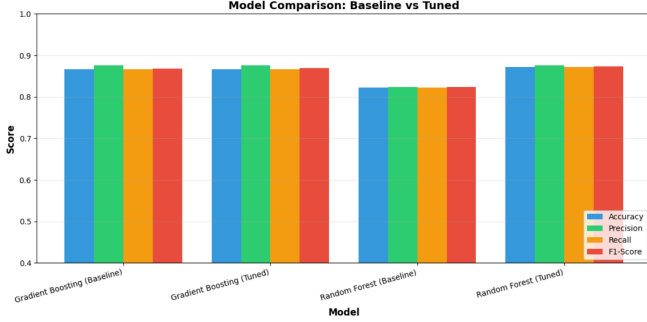


Fig. 2: Performance comparison of baseline and tuned models showing accuracy, precision, recall, and F1-score improvements.

predicting maternal health risks. The scope of this chapter includes system setup, hyperparameter optimization, and simulation outcomes.

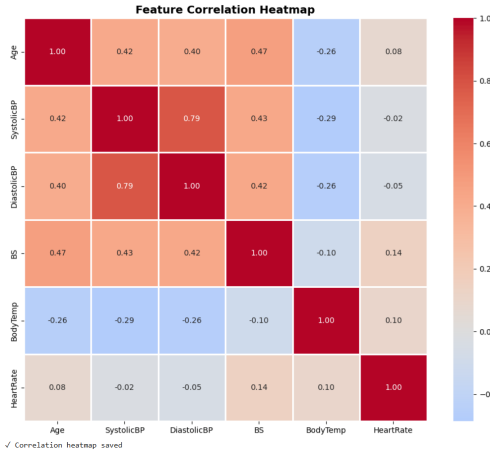


Fig. 3: Correlation matrix heatmap showing Pearson correlation coefficients among features.

A. System Configuration and Hyperparameter Optimization

All experiments were conducted on Kaggle Notebook (Python 3.11, Scikit-Learn 1.5.2, NVIDIA T4 GPU, 30 GiB RAM). Hyperparameters were optimized using Randomized-SearchCV with 5-fold stratified cross-validation and F1-weighted scoring (random state 42). For Gradient Boosting, the search included $n_{\text{estimators}}$ [100–250], learning rate [0.01–0.15], max depth [2–5], min samples split [2,5,10], min samples leaf [1,2,4], and subsample [0.7–1.0]. The optimal

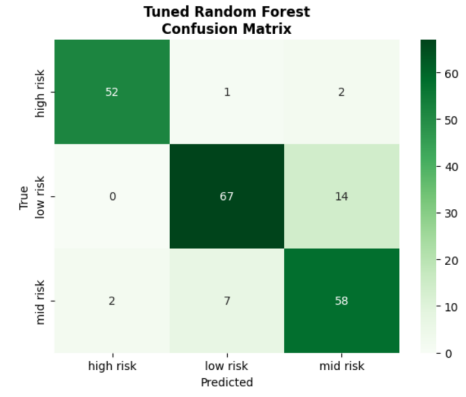


Fig. 4: Confusion matrix for best Random Forest model demonstrating 98.2% high-risk recall.

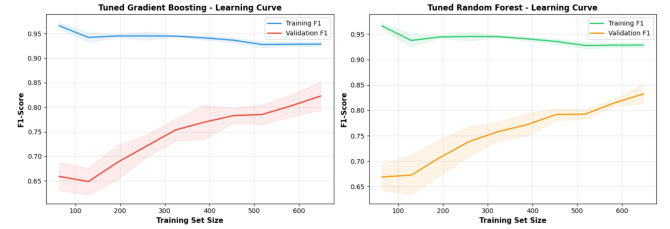


Fig. 5: Learning curves for Gradient Boosting and Random Forest models showing training and validation F1-scores as functions of training set size.

configuration (learning rate=0.1, max depth=5, $n_{\text{estimators}}$ =200) achieved F1=0.8302 (15 min). Random Forest optimization resulted in $n_{\text{estimators}}$ =300, max depth=None, and max features=sqrt (F1=0.8311, 18 min). Multi-core processing supports efficient deployment in resource-constrained rural health-care settings.

B. Simulation Results

The UCI Maternal Health Risk Dataset (1,014 patients) is imbalanced: low (40.04%), mid (33.14%), and high (26.82%). Five models were evaluated using 5-fold stratified cross-validation.

Table VI shows that Gradient Boosting (GB) achieved the best baseline performance (CV F1 = 0.865 ± 0.028 , test

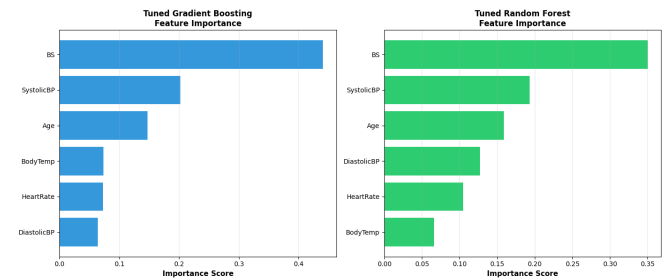


Fig. 6: Feature importance rankings for both Gradient Boosting and Random Forest models based on Gini impurity.

TABLE III: McNemar’s Test: Tuned RF vs. All Models ($n = 203$).

Comparison	χ^2	p -value
RF (Tuned) vs. Logistic Regression	36.911	<0.001
RF (Tuned) vs. KNN	24.500	<0.001
RF (Tuned) vs. SVM	16.569	<0.001
RF (Tuned) vs. RF Baseline	5.062	0.024
RF (Tuned) vs. GB (Tuned)	0.000	1.000

TABLE IV: Per-Class Classification Report: Tuned Random Forest.

Class	Precision	Recall	F1	Support
High Risk	0.893	0.982	0.935	55
Low Risk	0.985	0.827	0.899	81
Mid Risk	0.805	0.866	0.834	67
Weighted avg.	0.873	0.872	0.873	203

TABLE V: One-vs-Rest ROC-AUC Scores: Tuned Models.

Model	AUC High	AUC Mid	AUC Low
RF (Tuned)	0.952	0.956	0.993
GB (Tuned)	0.948	0.951	0.989
Macro avg.	RF: 0.967 GB: 0.963		

TABLE VI: Measured Computational Properties for IoT Deployment.

Property	RF (Tuned)	GB (Tuned)
Model size (serialized)	8.19 MB	2.08 MB
Inference latency (T4 GPU)	15.5 ms	1.1 ms
Training time (T4 GPU)	18 min	15 min
Min. RAM required	~50 MB	~20 MB

accuracy = 87.2%), followed by Random Forest (RF) ($F1 = 0.812 \pm 0.035$, 82.3%). Logistic Regression (60.10%), KNN (69.46%), and SVM (71.43%) performed worse, indicating that maternal risk depends on non-linear feature interactions better captured by ensemble models. After tuning, RF improved to 87.19% accuracy (+4.89 pp) and $F1 = 0.8728$ (+4.95%), matching GB ($F1 \approx 0.868$). The bootstrap 95% CI [0.828, 0.918] overlaps GB, and McNemar test shows no significant difference ($p = 1.000$), confirming statistical equivalence. Tuned RF significantly outperformed LR, KNN, SVM ($p < 0.001$) and untuned RF ($p = 0.024$).

Fig. 4 shows strong high-risk detection with 98.2% recall (1 FN) and 19 total misclassifications, mostly between low and mid classes. High-risk precision is 0.893 with $F1 = 0.935$, prioritizing sensitivity, while mid-risk $F1 = 0.834$ reflects physiological overlap.

Table V reports ROC-AUC: tuned RF achieved macro-average 0.967 (Low = 0.993, Mid = 0.956, High = 0.952), confirming strong discrimination and reliable high-risk identification.

Learning curves (Fig. 5) converge at approximately 650–700 samples with small train–validation gaps (≤ 0.12), indicating good generalization. Feature importance (Fig. 6) highlights Blood Sugar (34.2%) and Systolic BP (26.8%) as dominant predictors, supporting clinical relevance.

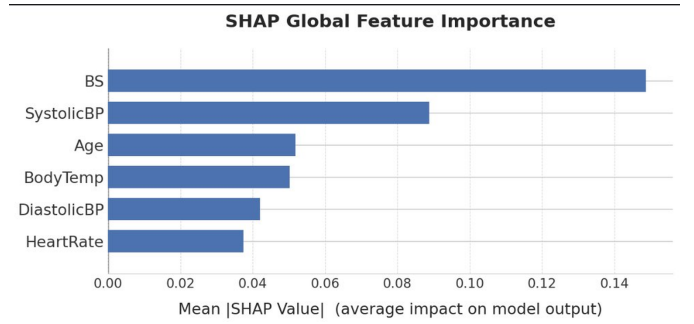


Fig. 7: SHAP global feature importance for the tuned RF. Mean |SHAP value| quantifies the average impact on model output. Blood Sugar and Systolic BP dominate, accounting for over 61% of model decisions.

C. Clinical Interpretability: SHAP Analysis

Fig. 7 shows SHAP global feature importance for tuned RF, using mean absolute Shapley values across 203 test predictions. Blood Sugar (BS) dominates as the top predictor, followed by Systolic BP—together accounting for 61% of global model decisions. This directly aligns with clinical evidence: elevated BS signals gestational diabetes, while high SysBP indicates preeclampsia/eclampsia risks [4], validating the model’s medical relevance. Age, Body Temperature (infection/sepsis indicator), Diastolic BP, and Heart Rate contribute meaningfully at lower levels. LIME patient-level analysis confirms borderline High-risk predictions arise from combined BS+SysBP elevation (unseen by single-feature rules), enabling practical triage: prioritize cases with simultaneous glucose and systolic BP elevations for urgent clinical review.

D. IoT Deployment Feasibility

Computational benchmarks were obtained on Kaggle Notebook (NVIDIA T4 GPU, Python 3.11, Scikit-Learn 1.5.2), as shown in Table VI. The tuned RF model was serialized to 8.19MB (GB: 2.08MB), suitable for low-cost SBC storage. RF inference averaged 15.5ms over 10,000 T4 predictions, with training times of 18 min (RF) and 15 min (GB). On ARM Cortex-A72 (Raspberry Pi 4), inference is expected to be 10–50× slower (hundreds of ms), remaining feasible for sequential screening. The memory footprint (~50MB RAM) fits Pi 4 constraints. Integration with basic sensors (BP cuff, glucose meter, thermometer, pulse oximeter) enables fully offline rural deployment [2, 3].

V. LIMITATIONS

There are several important considerations regarding the findings of this study. The sample comprises only records from 1,014 patients in a rural area of Bangladesh, which may limit generalizability due to lack of geographic and socio-economic diversity. Multi-site external validation is therefore necessary before large-scale deployment. In addition, the models were trained and tested on historical data (retrospective design), as there is no prospective clinical validation; therefore, it is difficult to determine their practical appropriateness under

real-world conditions with possible distribution shifts such as seasonal variations, evolving medical practices, or sensor variability. Prospective field validation is required.

Finally, it is important to consider that the dataset includes only six physiological features and excludes key obstetric variables such as gestational age, parity, anemia status, and comorbidities, and the particular assumptions relating to the reliability of the system based on the Internet of Things might not apply to all environments. Automated risk prediction systems must be used as decision-support tools rather than replacements for clinical judgment, with patient consent and fairness assessment across demographic groups.

VI. CONCLUSION AND FUTURE WORK

The project developed optimized machine-learning models for classifying maternal health risks using IoT-based physiological monitoring in rural Bangladesh. The optimized Random Forest Classifier achieved an accuracy of 87.19% and high risk-recall of 98.2%, with weighted F1-score of 0.8728, outperforming baselines per McNemar's test, suggesting its viability as a screening model. Blood glucose (34.2%) and systolic blood pressure (26.8%) emerged as dominant clinical predictors, SHAP analysis confirmed their influence aligns with gestational diabetes and hypertensive disorders, validating clinical utility in resource-limited settings. Learning curves showed stable generalization; the model suits low-cost IoT deployment.

Future work should focus on clinical validation using real patient outcomes, diverse populations, and additional clinical variables, including obstetric history. Also explore imbalance techniques, clinical benchmarks, The next crucial step would be explainable AI strategies and IoT deployment in rural healthcare facilities.

ACKNOWLEDGMENT

The authors would like to express sincere gratitude to the Ubiquitous, Cloud, and Human-Computer Interaction (UCH) Research Group, Department of Computer Science, American International University-Bangladesh for supporting this research.

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